Synthetic Data Generation for Skin Cancer Detection: Addressing Bias and Accessibility in Dermatological AI

Research Project Plan

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1. CONTEXT AND BACKGROUND

This project is part of the Research Group Sustainable Data & AI Applications at Fontys, which focuses on solving challenges when applying advanced artificial intelligence technology to medical imaging. Dermatological AI has evolved significantly since the early ISIC challenges (2016-2018), progressing from simple binary melanoma detection to complex multi-task analysis involving lesion segmentation, attribute detection, and disease classification across multiple diagnostic categories. However, the field faces persistent challenges in data collection, annotation costs, and representation bias that synthetic data generation could address. The project explores how semi-supervised learning with synthetic dermatological data can create more accessible, cost-efficient, and equitable diagnostic AI solutions for skin cancer detection.

1.1 RELATED WORK AND CURRENT STATE

The evolution of dermatological artificial intelligence has been closely tied to the development of increasingly sophisticated datasets and evaluation frameworks. The International Skin Imaging Collaboration (ISIC) challenge series represents a pivotal progression in the field, beginning with basic melanoma detection tasks using 900 images in 2016 (Gutman et al., 2016) and evolving into comprehensive multi-task analysis encompassing lesion segmentation, attribute detection, and disease classification with over 12,500 images by 2018 (Codella et al., 2018, 2019). This systematic expansion of challenge complexity reflects the growing sophistication of dermatological AI capabilities and the recognition that clinical practice requires more nuanced diagnostic support than simple binary classification.

Building upon these foundational challenges, Tschandl et al. (2018) introduced the landmark HAM10000 dataset, which fundamentally transformed the research landscape by providing 10,015 multi-source dermatoscopic images across seven distinct diagnostic categories. This dataset marked a crucial departure from the melanoma-nevus binary classification paradigm that had dominated earlier research, acknowledging the clinical reality that dermatologists must differentiate between multiple skin lesion types including melanoma, melanocytic nevi, basal cell carcinoma, actinic keratoses, benign keratosis, dermatofibroma, and vascular lesions. The multi-source nature of HAM10000, combining data from Austrian and Australian clinical settings, also introduced important considerations about dataset diversity and generalizability across different populations and clinical contexts.

Complementing these benchmark datasets, Combalia et al. (2019) contributed the BCN20000 dataset with 19,424 "dermoscopic lesions in the wild," specifically designed to address the challenge of unconstrained classification scenarios. This dataset's inclusion of hard-to-diagnose cases, nail lesions, and mucosal lesions highlighted the gap between controlled research environments and real-world clinical applications. Simultaneously, Rotemberg et al. (2021) revolutionized the field's approach to clinical context by introducing a patient-centric dataset containing 33,126 images from 2,056 patients, enabling research into contextual diagnostic strategies such as the "ugly duckling" sign, where unusual lesions are identified in comparison to a patient's other lesions.

Despite these advances, significant bias and representation issues have emerged as critical concerns in dermatological AI research. Groh et al. (2021) provided compelling evidence through their Fitzpatrick 17k dataset that dermatological AI models exhibit substantial performance disparities across different skin types, with models demonstrating significantly higher accuracy on skin types similar to their training data while systematically underperforming on darker skin types (Fitzpatrick types IV-VI). This finding has profound implications for healthcare equity, as it suggests that AI diagnostic tools may perpetuate or even exacerbate existing healthcare disparities. Furthermore, Raumanns et al. (2024) identified sex-based bias in CNN-based skin cancer detection systems, revealing higher diagnostic accuracy for male patients compared to female patients, thus highlighting additional dimensions of algorithmic bias that require attention.

These bias concerns are compounded by fundamental limitations in current dataset compositions, which predominantly feature melanocytic lesions and are heavily skewed toward European and light-skinned populations (Groh et al., 2021). This demographic imbalance not only limits the generalizability of trained models but also raises ethical questions about the equitable deployment of dermatological AI systems in diverse global populations. The convergence of these dataset limitations with the identified performance disparities creates an urgent need for more representative training data and bias mitigation strategies.

Current research gaps emerge at the intersection of these established challenges and emerging solutions. While synthetic data generation has shown promise in various domains, there remains limited integration of these techniques with established dermatological benchmarks such as HAM10000 and the ISIC challenge frameworks (Goodfellow et al., 2014; Radford et al., 2015). Existing approaches lack sufficient methods for generating representative synthetic data across diverse skin types and demographic groups, potentially perpetuating the same biases present in original datasets. Moreover, the field lacks comprehensive validation frameworks that can simultaneously assess synthetic data quality and clinical diagnostic relevance, making it difficult to ensure that synthetic augmentation strategies maintain clinical utility while addressing representation gaps. Finally, there is insufficient analysis of the cost-efficiency implications of synthetic data approaches compared to traditional annotation workflows, limiting the practical adoption of these potentially transformative technologies in resource-constrained healthcare settings.

2. PROBLEM STATEMENT

The problem statement section addresses the central challenge that motivates this research project: the intersection of data scarcity, privacy concerns, and equity issues in medical AI for dermatological applications. As healthcare systems worldwide increasingly rely on artificial intelligence for diagnostic support, the fundamental question of how to develop these systems ethically, efficiently, and equitably becomes paramount. This section establishes the specific challenges that synthetic data generation and semi-supervised learning aim to address within the broader context of dermatological AI development.

Primary Problem: Medical AI for skin cancer detection requires extensive labeled dermatological datasets, but medical data collection faces unique challenges: patient privacy concerns, rare disease representation, expensive expert annotation by dermatologists, and limited access to diverse patient populations. These barriers prevent the development of accessible and equitable diagnostic AI solutions.

Significance: Skin cancer is one of the most common forms of cancer worldwide, with early detection being crucial for patient outcomes. However, access to dermatological expertise is limited globally. AI-assisted diagnosis could democratize access to quality skin cancer screening, but current approaches require vast amounts of sensitive medical data, creating accessibility and privacy challenges.

Scope: This project focuses specifically on dermatological image analysis for skin cancer tissue detection, combining semi-supervised learning with synthetic medical data generation to create privacy-preserving, cost-efficient diagnostic AI solutions that comply with medical data regulations (GDPR, HIPAA considerations).

2.1 RESEARCH GAP

Despite significant advances in dermatological AI and semi-supervised learning, a critical gap exists at their intersection where synthetic data generation meets medical imaging requirements. Wang et al. (2024) and Zhang et al. (2025) represent the current state-of-the-art in SSL methodology, yet neither addresses the unique challenges of medical data authenticity and clinical safety requirements. This methodological gap is particularly pronounced given that Wang et al.'s RSMatch framework for synthetic data contamination detection remains untested in healthcare contexts where diagnostic accuracy directly impacts patient outcomes.

The bias and representation challenges identified by Groh et al. (2021) and Raumanns et al. (2024) reveal a second critical gap: while systematic biases in dermatological AI are well-documented, there is no established methodology for using synthetic data generation to actively mitigate these biases rather than perpetuate them. The underrepresentation of darker skin types and sex-based diagnostic disparities present clear targets for synthetic augmentation, yet current approaches lack validation frameworks that ensure both visual authenticity and clinical diagnostic relevance across diverse demographic groups.

Finally, the field lacks comprehensive validation and cost-efficiency frameworks that bridge research advances with practical healthcare deployment. While Raumanns et al.'s ENHANCE dataset demonstrates improved performance through multi-task learning (AUC: 0.811 vs 0.794), the integration of such approaches with synthetic data workflows remains unexplored. More critically, there is insufficient analysis of whether computationally intensive SSL methods can compete with traditional dermatologist annotation workflows in terms of cost-effectiveness, particularly when regulatory requirements and clinical safety standards are considered. This gap between methodological sophistication and practical healthcare implementation represents the most significant barrier to translating SSL-synthetic data advances into accessible diagnostic solutions.

Thus, the specific research gap that the project is going to explore is, that there is no established methodology for effectively integrating semi-supervised learning with synthetic data generation specifically for dermatological applications that simultaneously addresses bias mitigation, maintains clinical diagnostic accuracy, ensures cost-efficiency compared to traditional annotation workflows, and provides comprehensive validation frameworks suitable for healthcare deployment. This research gap represents the critical missing link between advanced SSL methodologies and practical, equitable dermatological AI solutions.

3. RESEARCH QUESTIONS

The research questions section translates the identified research gap into specific, actionable inquiries that will guide the systematic investigation of synthetic data generation and semi-supervised learning applications in dermatological AI. These questions are structured hierarchically, with a main research question addressing the overarching challenge and three sub-questions that decompose the problem into manageable components focusing on data generation, system implementation, and validation respectively. This structured approach ensures comprehensive coverage of the technical, economic, and clinical validation aspects necessary for developing practically deployable healthcare AI solutions.

Main Research Question

How can semi-supervised learning and synthetic data help to achieve cost-efficient and sustainable AI software solutions for skin cancer detection?

Sub-questions

1. How to create a cost-efficient solution for generating and labeling synthetic dermatological data for skin cancer detection?
2. How to create a semi-supervised learning environment and deployment workflow with synthetic dermatological data?
3. How to validate the quality of semi-supervised learning models and synthetic dermatological data, and assess overall cost and performance?

4. RESEARCH METHODS

The research methods section outlines a comprehensive mixed-methods approach that combines quantitative and qualitative research techniques to address each sub-research question systematically. The methodology integrates literature studies, expert consultations, exploratory data analysis, prototyping, architectural design, multi-criteria decision making, and rigorous validation frameworks. This multi-faceted approach ensures that the research addresses both technical feasibility and clinical relevance while maintaining scientific rigor and practical applicability. Each method is specifically selected to contribute unique insights that collectively build toward answering the main research question about achieving cost-efficient and sustainable AI solutions for skin cancer detection.

Sub Question	Research Methods	
	Method	*Description*
How to create a cost-efficient solution for generating and labeling synthetic dermatological data for skin cancer detection?	**Literature study**	**What:** A systematic review of research and grey literature on skin-lesion imaging, synthetic image generation, weak or automated labeling, and fairness in dermatology AI. **Why:** To ground targets and constraints in existing evidence, avoid known pitfalls such as artifact bias, and set realistic goals for cost and quality. **How:** Define search strings and sources, apply inclusion and exclusion criteria during screening, extract salient findings on lesion mix, skin-tone coverage, and common failure modes, then synthesize these insights into a brief that guides synthesis targets and labeling rules.
	Expert interview	**What:** Interview experts, covering workflows, label definitions, typical artifacts, and tolerable levels of label noise. **Why:** To translate clinical practice into practical generation prompts, a workable label schema, and review processes that control cost without sacrificing quality. **How:** Draft an interview guide, run focused sessions of thirty to forty-five minutes, code and validate key insights with participants, then convert the results into labeling heuristics and standard operating procedures.
	Exploratory data analysis	**What:** Descriptive analysis of the available real datasets to understand class balance, estimated skin-tone distribution, artifact frequency, image resolution, and missingness. **Why:** To identify where synthetic data and weak labels are most needed and to set synthesis quotas that correct under-representation. **How:** Compute summary statistics and visualizations, estimate subgroup distributions such as Fitzpatrick groups, detect duplicates and artifacts, and compile the gaps that will inform synthesis ratios and labeling priorities.

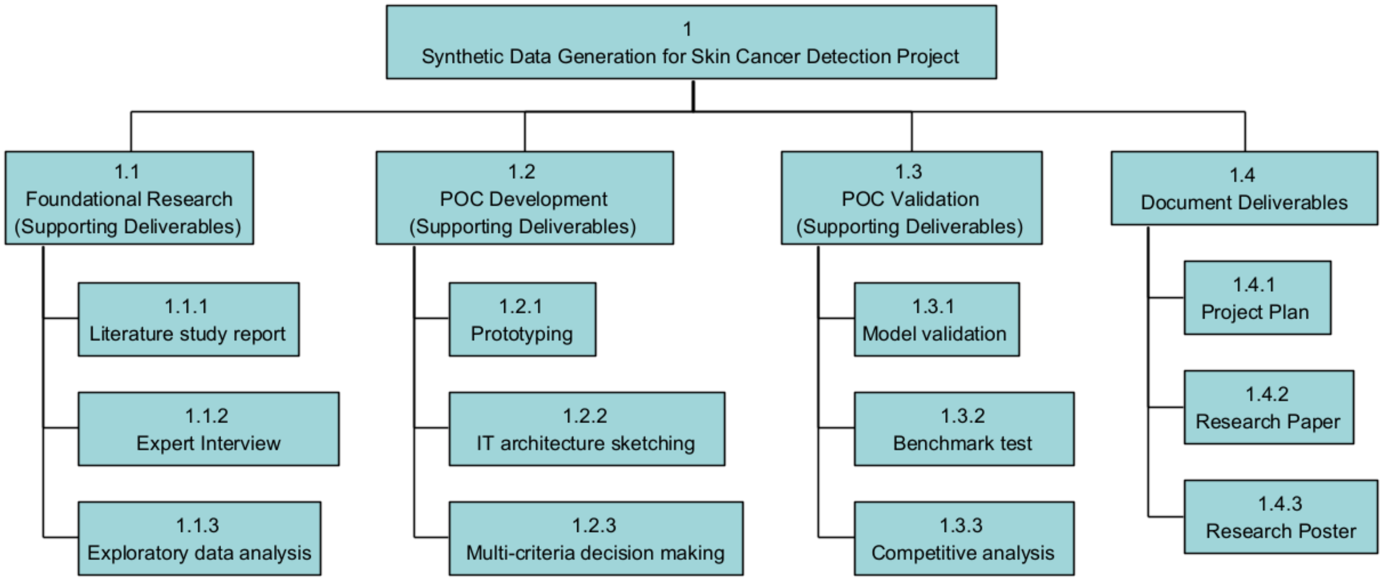
Sub Question	**Research Methods**	
	Method	*Description*
How to create a semi-supervised learning environment and deployment workflow with synthetic dermatological data?	**Prototyping**	**What:** Rapid, iterative builds of the semi-supervised pipeline that combines real and synthetic data, with a focus on learning and risk reduction. **Why:** To validate key assumptions early, such as synthetic-to-real ratios, loss weights, and provenance handling, before investing in a full production stack. **How:** Set explicit learning goals, implement a minimal pipeline that includes data loaders with origin flags and a ratio scheduler, then run short experiments, keep what works, and discard what does not.
	IT architecture sketching	**What:** A high-level sketch of the training and deployment architecture that traces data ingress, preprocessing, semi-supervised training, evaluation, model registry, continuous integration and delivery, inference, monitoring, and governance controls. **Why:** To align stakeholders quickly, reveal integration and compliance risks, and clarify responsibilities and control points. **How:** Convene a small group of stakeholders, sketch the end-to-end flow with roles and checkpoints such as a data protection assessment and model cards, capture the sketch, and translate it into a concise diagram that guides implementation.
	Multi-criteria decision making	**What:** A structured comparison of design options such as model backbones, semi-supervised variants, and infrastructure choices, evaluated against weighted criteria. **Why:** To make transparent and evidence-based technology choices that balance accuracy, fairness, latency, operability, compliance, and cost. **How:** Define candidate options, co-create criteria and weights with stakeholders, score options using evidence from prototypes and benchmarks, examine the sensitivity of outcomes to the weights, and select the leading candidate with a clear decision rationale.

Sub Question	**Research Methods**	
	Method	*Description*
How to validate the quality of semi-supervised learning models and synthetic dermatological data, and assess overall cost and performance?	**Model validation**	**What:** Statistical validation of the semi-supervised model using locked, real-only test data and predefined acceptance thresholds. **Why:** To ensure results are trustworthy, generalize beyond the training distribution, and meet clinically acceptable performance levels. **How:** Fix train, validation, and test splits, predefine metrics such as ROC-AUC, sensitivity at ninety percent specificity, precision-recall AUC, and calibration error, apply cross-validation or bootstrapping and appropriate statistical tests, and document pass or fail outcomes against evaluation thresholds.
	Benchmark test	**What:** A fair and repeatable comparison between the best candidate model and baseline models using a standardized benchmark or a fixed external test set. **Why:** To demonstrate progress beyond simple baselines and show robustness under identical evaluation conditions. **How:** Select a recognized benchmark or lock an external test set, define a fixed evaluation protocol and random seeds, run baselines and candidates under the same conditions, and report comparative tables and plots with clear ranking.
	Competitive analysis	**What:** A structured scan of competing academic and industry models, datasets, and tool chains relevant to skin-lesion detection. **Why:** To position the proposed solution, highlight differentiators such as performance on darker skin tones or strong governance practices, and identify gaps to address next. **How:** List relevant competitors, collect publicly available claims on performance, data diversity, deployment features, and governance practices, build a comparison grid and a concise strengths and weaknesses summary, and turn the findings into concrete priorities for improvement and communication.

5. PROJECT DELIVERABLES

The project deliverables section outlines the comprehensive set of tangible outputs that will be produced throughout this research project. These deliverables are strategically designed to ensure both academic rigor and practical applicability, supporting knowledge transfer to the research community while enabling future implementation in clinical settings. The deliverables encompass academic publications, proof of concept artifacts, technical documentation that collectively demonstrate the project's contributions to advancing cost-efficient and equitable dermatological AI solutions.

5.2 PRODUCT BREAKDOWN STRUCTURE



6. STAKEHOLDER OVERVIEW

The stakeholder overview section identifies and analyzes the key individuals and groups who have vested interests in this research project's success and outcomes. Effective stakeholder management is crucial for research projects in medical AI, where technical innovation must align with clinical needs, regulatory requirements, and academic standards. This section establishes clear roles, responsibilities, and engagement strategies for each stakeholder, ensuring that the project benefits from diverse expertise while maintaining accountability and communication throughout the research process. The collaborative approach outlined here enables the integration of technical innovation with clinical relevance and academic rigor.

Stakeholder	Role	Responsibilities	Interest	Involvement
Qin Zhao (q.zhao@fontys.nl) [Project Owner]	Project supervisor and primary stakeholder	Project guidance, oversight and evaluation	Ensuring research quality, methodological rigor, and successful project completion	Regular supervision meetings, milestone reviews, and academic mentorship
Ralf Raumanns (ralf.raumanns@fontys.nl) [Domain Expert]	Skin cancer detection specialist and dermatological AI researcher	Provide expert guidance and support validation of the results	Advancing bias-aware dermatological AI and improving diagnostic equity	Domain expertise consultation, research methodology guidance, and validation of medical AI approaches

7. PLANNING

The planning section provides a comprehensive roadmap for executing this research project within a structured 16-week timeframe. Effective project planning is essential for complex research endeavors that integrate multiple methodologies, stakeholder requirements, and deliverable targets. This section establishes clear phases, milestones, and deadlines that ensure systematic progress from initial literature review through experimental validation to final documentation and knowledge transfer. The planning framework balances thorough research investigation with practical constraints, enabling the delivery of high-quality research outcomes that advance both academic knowledge and practical applications in dermatological AI.

7.1 PROJECT TIMELINE

Phase	Description
Phase 1: Literature Review and Benchmark Analysis [Weeks 1-3]	• Comprehensive analysis of ISIC challenge evolution (2016-2018) and task progression • Dataset characterization: HAM10000 7-category analysis, BCN20000 "in-the-wild" challenges, Fitzpatrick 17k bias patterns • Review of Raumanns et al.'s multi-task learning and crowdsourced annotation methodologies • Analysis of patient-centric approaches from Rotemberg et al.'s clinical context integration • Semi-supervised learning and synthetic data generation literature review
Phase 2: Solution Design and Prototype Development [Weeks 4-8]	• Design synthetic data generation pipeline (API-based approach vs. fine-tuning evaluation) • Develop semi-automated annotation workflow using baseline/weakly-trained models • Create end-to-end pipeline (data generation to model evaluation) • Implement quality filtering system for "hallucination" detection in synthetic data • Prompt engineering optimization for domain-specific generation
Phase 3: Experimental Implementation and Validation [Weeks 9-12]	• Conduct systematic experiments with multiple dataset compositions (real-only, synthetic-only, hybrid ratios) • Implement comprehensive validation using COCO metrics, confusion matrices, and domain gap analysis • Measure annotation time reduction and cost-efficiency metrics • Validate model collapse prevention through hybrid training strategies • GDPR compliance assessment for synthetic data generation
Phase 4: Documentation and Reporting [Weeks 13-16]	• Research paper writing • Technical documentation for handover • Research poster preparation • Code quality assurance and documentation

7.2 KEY MILESTONES AND DEADLINES

Critical Milestone	Completion time
Literature review completion	Week 3
Prototype proof-of-concept completion	Week 8
Validation framework implemented	Week 10
Experimental results and analysis	Week 12
First draft of final paper	Week 14
Technical documentation completion	Week 15
Final paper completion	Week 16
Project poster completion and presentation	Week 16
Project handover	Week 16

8. REFERENCES

Codella, N., Gutman, D., Celebi, M., Helba, B., Marchetti, M., Dusza, S., . . . Halpern, A. (2018). Skin lesion analysis toward melanoma detection: A challenge at the 2017 international symposium on biomedical imaging (ISBI), hosted by the international skin imaging collaboration (ISIC). Retrieved from https://arxiv.org/abs/1710.05006

Codella, N., Rotemberg, V., Tschandl, P., Celebi, M. E., Dusza, S., Gutman, D., . . . Halpern, A. (2019). Skin Lesion Analysis Toward Melanoma Detection 2018: A Challenge Hosted by the International Skin Imaging Collaboration (ISIC). Retrieved from https://arxiv.org/abs/1902.03368

Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., . . . Bengio, Y. (2014). Generative Adversarial Networks. *Proceedings of the 28th International Conference on Neural Information Processing Systems - Volume 2* (pp. 2672–2680). Cambridge, MA, USA: MIT Press. Retrieved from https://arxiv.org/abs/1406.2661

Groh, M., Harris, C., Soenksen, L., Lau, F., Han, R., Kim, A., . . . Badri, O. (2021). Evaluating Deep Neural Networks Trained on Clinical Images in Dermatology with the Fitzpatrick 17k Dataset. *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)* (pp. 1820-1828). Nashville: IEEE. doi:10.1109/CVPRW53098.2021.00201

Gutman, D., Codella, N., Celebi, E., Helba, B., Marchetti, M., Mishra, N., & Halpern, A. (2016). Skin Lesion Analysis toward Melanoma Detection: A Challenge at the International Symposium on Biomedical Imaging (ISBI) 2016, hosted by the International Skin Imaging Collaboration (ISIC). Retrieved from https://arxiv.org/abs/1605.01397

Radford, A., Metz, L., & Chintala, S. (2015). Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks. *CoRR*. Retrieved from https://doi.org/10.48550/arXiv.1511.06434

Raumanns, R., Schouten, G., Joosten, M., Pluim, J. P., & Cheplygina, V. (2021). ENHANCE (ENriching Health data by ANnotations of Crowd and Experts): A case study for skin lesion classification. *Machine Learning for Biomedical Imaging*, 1-26. doi:https://doi.org/10.59275/j.melba.2021-geb9

Raumanns, R., Schouten, G., Pluim, J. P., & Cheplygina, V. (2024). Dataset Distribution Impacts Model Fairness: Single Vs. Multi-task Learning. *Ethics and Fairness in Medical Imaging* (pp. 14-23). Springer, Cham. doi:https://doi.org/10.1007/978-3-031-72787-0_2

Rotemberg, V., Kurtansky, N., Betz-Stablein, B., Caffery, L., Chousakos, E., Codella, N., . . . Nanda, J. (2021). A patient-centric dataset of images and metadata for identifying melanomas using clinical context. *Scientific Data*, 34. Retrieved from https://doi.org/10.1038/s41597-021-00815-z

Tschandl, P., Rosendahl, C., & Kittler, H. (2018). The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data, 5*(1). Retrieved from https://doi.org/10.1038/sdata.2018.161

Wang, Z., Mao, J., Xiang, L., & Yamasaki, T. (2024). From Obstacles to Resources: Semi-supervised Learning Faces Synthetic Data Contamination. Retrieved from https://arxiv.org/abs/2405.16930

Zhang, P., Mai, Z., Nguyen, Q. H., & Chao, W. L. (2025). Revisiting semi-supervised learning in the era of foundation models. Retrieved from https://arxiv.org/abs/2503.09707